

Excess Self-Confirmation Is Domain-Dependent: Self-Preference Drives It in Code, the Optimizer’s Curse Alone Explains Math

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Abstract

When a language model generates candidates, judges them, and revises under its own scores, reported progress can outrun true progress — mostly the optimizer’s curse (argmax over a noisy judge inflates the selected score; true quality is unaffected), not self-deception. We decompose the raw gap into a selection term (SEL) and a residual, ESC (“excess self-confirmation”), and ask on a real closed loop (Qwen3.5-4B, MLX, HumanEval+MBPP, GSM8K+MATH, $N=4$, $T=15$, 84–91 items/domain) whether ESC reflects self-preference or generic judge effects. **We find a clean, replicated domain split, not a general answer.** In code, self-judgment shows excess self-confirmation on **four independent statistics** — $ESC(T) = +0.115$ ($p=0.0013$), cumulative $AUC_{ESC} = +1.76$, a noise-matched differential $ESC_{sp} = +0.129$ ($p=0.012$), and a placebo-adjusted differential $ESC_{adj} = +0.177$ ($p=0.002$) — and code’s raw gap is a coincidental cancellation of two significant, opposite-signed terms ($\Delta SEL = -0.114$, $ESC = +0.115$) a raw-gap monitor would miss. In math, none of the same five statistics confirms excess self-confirmation, and two are *significantly the wrong sign*: math’s self-judge is *better calibrated* than a content-free placebo judge once revision-conditioning inflation is subtracted out. A methodological finding accompanies this: three unrelated strong judges (12B local, frontier API, reasoning-tuned MoE) all fail the same noise-matching gate on math, converging on nearly the same operating point — math’s judge saturation is structural, not one judge’s artifact. Full harness, trajectory library, and code: github.com/deepgrounding/excess-self-confirmation. Design frozen pre-data-collection (tags `prereg-opt2-v2`, `prereg-opt2-v2-noise`); amendments disclosed in full.

1. Introduction

Self-evaluation loops — generate N candidates, score them with a judge, select the argmax, feed the judge’s feedback into the next revision — are now a standard inference-time and agentic pattern (Madaan et al. 2023; Shinn et al. 2023; Yuan et al. 2024). When the same model (or a close relative) plays both generator and judge, the loop’s *reported* progress routinely exceeds its *true* progress. That fact is not news. It is the familiar shape of Goodhart’s law and of reward-model overoptimization (Gao et al. 2023; Manheim and Garrabrant 2018): optimize a noisy proxy hard enough and the proxy rises while the quantity you care about stalls or falls.

What *is* less obvious, and what this paper isolates empirically, is that a large and statistically inevitable piece of the reported–true gap is **not** self-preference, circular feedback, or rubric hacking.

It is the **optimizer’s curse** (Smith and Winkler 2006; Thaler 1988): taking the argmax of N noisy estimates systematically inflates the selected estimate relative to the population mean, while the selected item’s *true* quality is not inflated by the same amount. A raw gap that includes this term — call it the raw self-confirmation gap, raw SCG — is almost guaranteed to be positive under any noisy best-of- N protocol. Treating that positive gap as evidence of “self-confirming drift” is a category error: it confuses a selection artifact with a closed-loop phenomenon. Once that baseline is subtracted, what remains is not a single number either — it is domain-dependent, and this paper’s central contribution is showing exactly how and why.

A recent survey of recursive self-improvement (Chen et al. 2026) names the evaluator as the public bottleneck of self-improvement, lists self-confirming loops and diversity collapse among the characteristic failure modes, and flags governance-grade measurement — audit and early-stop warning — as the most underdeveloped niche. Concurrent work in the same series (Chen 2026) *manipulates* evaluator-generator error correlation ρ under fixed marginal accuracy and asks when refinement flips from net-positive to net-negative. The present paper takes the complementary observational cut: in a *real* closed loop, how much of the reported-true gap is the optimizer’s curse, how much is excess drift beyond that curse, and how much of the excess is attributable to self-preference specifically, once every generic confound — selection alone, revision-conditioning style inflation, and judge noise — is controlled for.

What this paper is. We pre-registered three hypotheses (H1–H3; §3), a metric decomposition and an early-warning protocol (§4–5), and validated the instrument on synthetic sandboxes with known ground truth (§5.4) before touching real models. We then executed the real-model arm for H1 and H2 to completion on cell [A]’s reduced scope (§6–7; see §8 for exactly what was and was not run, and why); H3’s collapse-induction cells remain future work (§9). This is a **findings paper built on a registered design**, not a two-stage registered report: the original plan was to split design (Stage 1) and results (Stage 2) across two arXiv submissions, but with H1/H2’s real-model arm now complete, we report the full picture — pre-registered design, deviations, and results — in one document, anchored by public git tags rather than a separate design-only submission (§8 explains this and every other amendment openly).

Contributions. (1) An algebraic decomposition $\text{rawSCG} = \Delta\text{SEL} + \text{ESC}$ with the identity $\text{ESC}(t) = \text{bias}_{\text{pop}}(t) - \text{bias}_{\text{pop}}(0)$ — excess confirmation equals candidate-pool optimism drift, so per-round winner’s curse is removed by construction, validated to machine precision on synthetic sandboxes and on every real trajectory pool in this paper. (2) A real-model result, replicated across four independent statistical comparisons per domain, that excess self-confirmation is present and self-preference-attributable in code and consistently absent (twice significantly the wrong sign) in math — not a power artifact, since the placebo and noise-matched controls that resolve it are themselves independently significant. (3) A pre-registered contingency (comparison against a content-free placebo judge) that fires in math and is not a null result: it shows math’s self-judge is *better calibrated* than a judge with no information at all, once generic revision-conditioning inflation is removed. (4) A methodological finding that judge-score saturation on math generalizes across three structurally unrelated strong judges, evidence that it is a property of the grading task rather than any one model. (5) A within-condition early-warning protocol (H3), specified and validated on synthetic data but not yet run on real collapse data — reported as a validated instrument and an open real-model question, not a result. (6) A complete, openly-amended pre-registration and harness runnable on a single Apple Silicon Mac Mini, with code and the full candidate-pool trajectory library released.

2. Related Work

Self-refinement and its limits. Self-Refine (Madaan et al. 2023) and Reflexion (Shinn et al. 2023) established iterative critique-and-revise; Self-Rewarding LMs (Yuan et al. 2024) and STaR (Zelikman et al. 2022) moved related loops into training. Huang et al. (Huang et al. 2024) showed that LLMs cannot reliably self-correct reasoning without external feedback; Kamoi et al. (Kamoi et al. 2024) survey when self-correction actually works. Our $N=1$ arm is the Reflexion-style pure-revision path; its SCG *is* ESC, and any positive value there cannot be blamed on best-of- N selection — in our data, it is not positive in either domain (§7).

Reward overoptimization and the optimizer’s curse. Gao, Schulman, and Hilton (Gao et al. 2023) document the proxy–true reward gap under KL-constrained optimization against a learned reward model. We treat their gap as isomorphic to our *raw* SCG and explicitly decline to claim it as a finding on its own. What we add is a decomposition into a per-round selection term (SEL; the optimizer’s / winner’s curse of (Smith and Winkler 2006; Thaler 1988)) and a residual pool-drift term (ESC), and an empirical answer for when the residual is attributable to self-preference. Reward-model ensembling (Coste et al. 2024; Eisenstein et al. 2023) reduces overoptimization by de-correlating proxy errors; our ESC_{sp} comparison asks how much residual gap remains after *noise* (not correlation structure) is matched between the self-judge and an external one.

Self-preference and judge bias. MT-Bench documents position, verbosity, and self-enhancement biases in LLM-as-judge (Zheng et al. 2023); Panickssery et al. show self-preference is driven by self-*recognition* (Panickssery et al. 2024); length biases are a known confound (Dubois et al. 2024). These are mostly *static* scoring results. We push self-preference into a *closed loop* and ask whether the bias compounds across rounds as $\text{ESC}_{\text{sp}}(t)$, after noise matching has removed the “stronger judge = less winner’s curse” confound — and we find it does, but only in one of two domains tested, which the static-scoring literature has no mechanism to predict.

Self-confirming loops, collapse, and diversity. Tan et al. (Tan et al. 2025) diagnose systemic reward bias in self-rewarding RL. Model collapse under recursive training on self-generated data (Shumailov et al. 2024) and diversity collapse (distinct- n , self-BLEU; (Li et al. 2016; Zhu et al. 2018)) are adjacent failure modes. Our design includes an arbitrable-rubric, high-selection-pressure collapse-induction cell intended to test whether early ESC slope predicts late collapse within a fixed condition (H3); that cell was not run at real-model scale in this paper (§9) and is reported as a validated-instrument, open-question future direction rather than a result.

Companion measurement paper. Chen (Chen 2026) constructs a Gaussian-copula synthetic evaluator that sweeps error-structure ρ at fixed (FPR, FNR) and derives a closed-form refinement surface. That paper *manipulates* evaluator error structure; this one *observes* closed-loop confirmation drift on real trajectories. Shared harness, trajectory schema, and $\hat{\rho}$ estimator are intentional (§11).

Positioning. We do not claim to discover that proxy scores can diverge from truth. We claim a clean subtraction of the optimizer’s-curse baseline inside real self-evaluation loops, a domain-dependent answer (not a single number) for how much of the residual is self-preference-attributable, and a methodological demonstration that this asymmetry is not a statistical-power artifact — it survives four independent, mutually reinforcing comparisons per domain.

3. Problem Setup and Pre-Registered Hypotheses

3.1 Research questions

Closed-loop self-evaluation: a model samples candidates, a judge scores them, argmax selection plus feedback drives the next round. When the model is both producer and judge, reported progress can detach from true progress — but much of the detachment is the optimizer’s curse. We asked, before collecting real-model data:

- **RQ1 (excess existence).** After subtracting the per-round optimizer’s-curse baseline, does a positive excess ESC remain? The $N=1$ arm (no selection) is the cleanest path: any gap there cannot be winner’s curse.
- **RQ2 (self-preference, not just noise).** After matching a strong external judge’s discrimination (AUC) to the self-judge, does a positive ESC_{sp} remain?
- **RQ3 (within-condition early warning).** Inside a fixed (model, task, judge, rubric) condition, does early ESC slope predict late true-score collapse across difficulty-bin $\times$ cluster $\times$ seed trajectories — after beating a condition-label baseline and a within-condition permutation null? (*Validated on synthetic data only in this paper; §9.*)

3.2 Hypotheses

H1 (excess exists; not just Goodhart). On verifiable tasks, under the self-judge,

$$ESC(t) = raw_SCG(t) - (SEL(t) - SEL(0))$$

terminal $ESC(T)$ and cumulative AUC_{ESC} are significantly > 0 (one-sided, task-level cluster bootstrap); *and* $SCG_{N=1}(T)$ is significantly > 0 on the pure-revision arm. *Falsified if* both paths fail to reject the null — a clean, publishable negative: under this setup the self-confirmation gap *is* the optimizer’s curse. **Result (§7): confirmed in code (4/4 relevant statistics, plus consistent cell-[B] direction is the only miss), not confirmed in math (0/5).**

H2 (self-preference excess; not just noise). After noise matching so $AUC(J_{strong\sim}) \approx AUC(J_{self})$ with empirical SEL magnitudes in ratio $[0.8, 1.25]$,

$$ESC_{sp}(t) = ESC_{self}(t) - ESC_{strong\sim}(t)$$

is significantly > 0 . *Falsified if* the CI covers 0 — then the verification-hierarchy ranking self $<$ peer $<$ strong is fully explained by noise / winner’s-curse magnitude, and governance reduces to lowering judge variance. **Result (§7): confirmed in code ($p=0.012$); math’s dual-bound estimate is the wrong sign under both bounds ($p=0.95$, $p=0.93$).**

H3 (within-condition early warning; validated as an instrument, not yet tested on real collapse data). With trajectory unit = (difficulty bin $\times$ item cluster $\times$ seed) inside a fixed (model, domain, judge, rubric, N) cell, early-window ($k=3$) ESC slope predicts late collapse:

- within-condition warning AUC significantly > 0.5 ;
- significantly above a condition-label-only logistic baseline and above the 95th percentile of a within-condition label-permutation null;
- above the AUC obtained by replacing the feature with early slope of *raw* SCG (incremental value of the decomposition).

Falsified if within-condition AUC is consistent with chance or fails to beat the label / permutation baselines — then warning power is only the trivial reflection of condition tags. **Status:**

the statistic and its defenses against the label confound are validated end-to-end on synthetic data with a known ground-truth collapse mechanism (§5.4, Check 5); the collapse-induction cells needed to test H3 on real trajectories were not run in this paper (§9).

Each failure mode is an informative governance conclusion, not a null paper (§9).

4. Method

4.1 Closed-loop harness

```
for t = 0, 1, ..., T:
  1. Generate:
    t=0: sample N independent baseline solutions C_0 (no history)
    t>=1: condition on s_{t-1} + judge feedback; sample N revisions C_t
  2. Score: judge outputs j(c_i) in [0,1] + feedback for the eventual selectee
  3. Select: s_t = argmax_i j(c_i)      # N=1: the sole candidate
  4. Offline (judge never sees): oracle G(c_i) for EVERY candidate in C_t
    -> J_sel, G_sel, J_pop, G_pop every round
    -> SEL, ESC, analytic baseline, random-select counterfactuals: all offline
```

Three structural choices relative to a naive gap protocol: (i) round 0 also draws a full pool of N , so SEL/ESC windows are comparable across rounds; (ii) the oracle scores the *entire* pool, which is the prerequisite for measuring winner’s curse; (iii) an explicit $N=1$ arm and an arbitrable-rubric condition are first-class cells, not afterthoughts.

Isolation constraints. Generation side is frozen across judge conditions (same model, decoding, seeds, revision template); only who scores, how selection proceeds, and what feedback is written may vary. The verifier never enters the judge’s context. Judge outputs are constrained to a shared $[0, 1]$ correctness scale under a shared rubric skeleton (faithful or arbitrable).

4.2 Models

Local inference via MLX (mlx-1m), 4-bit throughout, except the strong-judge cross-checks (below), which run via API/local-server.

Role	Model	Use
Generator, cheap scale point	Qwen3.5-0.8B (non-thinking)	Harness pilot only (§10.1); no cell [E] real-model arm was run
Generator, primary	Qwen3.5-4B (non-thinking)	Full H1/H2 matrix (this paper’s real-model results)
Generator, large	Qwen3.5-9B (non-thinking)	Reserved for H3 collapse induction (not run; §9)
Peer judge J_{peer}	Llama-3.2-3B-Instruct	Same size class, different family
Strong judge J_{strong} , primary	Gemma-4 12B, local via Ollama	Default; noise-matched for H2
Strong judge, cross-checks	Claude Haiku 4.5 (API); DeepSeek V4 Flash 0731 (API, math only)	Independent replications of the math judge-saturation finding

Two amendments to the original design, both made before any results were collected and both documented in full in §8:

Strong judge (pre-data-collection amendment). The original design anchored J_{strong} on Claude Haiku 4.5 via API, with a local 14B/32B model as an outage contingency. We switched the default to a fully local judge — Gemma-4 12B via Ollama — for zero marginal cost and full reproducibility without an API key, before the noise-matching calibration (Appendix B) was run for either domain; Claude Haiku 4.5 is retained as an independent cross-check rather than dropped, and both domains’ calibrations were run against both judges (§7).

Third judge, math only (post-hoc addition, reported regardless of outcome). After Gemma and Haiku both failed the same noise-matching gate on math (§7), we added a third, architecturally unrelated judge — DeepSeek V4 Flash 0731, a reasoning-tuned sparse-MoE model, via OpenRouter — specifically to test whether the gate failure is judge-specific or structural to math grading. This is exploratory by construction (it was not in the original design) and is reported as such; §7 gives the result.

Model family substitution from the original proposal draft (Qwen2.5-Instruct) to Qwen3.5, matching the companion registered report (Chen 2026) under the same size-class-preserving, non-thinking-mode contingency, changed no hypothesis or grid — only the checkpoint family (Qwen Team 2026, 2025).

4.3 Tasks, ground truth, trajectory units

- **Code:** HumanEval + sanitized MBPP (Chen et al. 2021; Austin et al. 2021); verifier = hidden unit-test pass rate.
- **Math:** GSM8K + MATH subset (Cobbe et al. 2021; Hendrycks et al. 2021); verifier = normalized exact match.
- **Screening:** items with baseline pass rate in $[0.25, 0.75]$ (widened from the original $[0.3, 0.6]$; §8); frozen pools of **84 math** and **91 code** items (§10.2 details the screen).
- **H3 units (specified, not yet used on real data):** tertile difficulty bins from round-0 pool true scores; two embedding clusters per bin; trajectory = (bin $\times$ cluster $\times$ seed) averaging 8–17 items. Power target: ≥ 12 trajectories per H3 condition, ≥ 8 items each.

4.4 Judge conditions

Unified interface `score(task, candidate) -> (p_correct in [0,1], feedback_text):`

- **Primary:** J_{self} (generator judges itself); J_{peer} ; J_{strong} .
- **Derived (H2 core):** $J_{\text{strong}\sim} - J_{\text{strong}}$ scores plus frozen calibrated noise so AUC matches J_{self} . Noise amplitude is calibrated *offline* on the round-0 pool (Appendix B); the arm itself runs *online* (judge score $\rightarrow$ inject frozen noise $\rightarrow$ argmax $\rightarrow$ use the selectee’s J_{strong} feedback). Offline reconstruction of the full arm is impossible once selection diverges.
- **Controls:** J_{oracle} (verifier as judge; $\text{SEL} \equiv 0$, $\text{ESC} \equiv 0$ by algebraic construction — an instrument check, not a result); J_{placebo} (random scores + length-matched uninformative feedback). Placebo has two jobs: (i) instrument zero — raw SCG may be large via winner’s curse but ESC should be ≈ 0 ; (ii) revision-conditioning length/style inflation — if $\text{ESC}_{\text{placebo}}$ is significantly > 0 , the pre-registered conditional analysis reports $\text{ESC}_{\text{adj}} = \text{ESC} - \text{ESC}_{\text{placebo}}$. **This contingency fired in math (§7).**

4.5 Selection pressure and arbitrable rubric

$N \in \{1, 4, 8, 12\}$: $N=1$ is pure revision ($\text{SEL} \equiv 0$; run at full scale, both domains, in this paper); $N=4$ is the core matrix (run at full scale, both domains); $N \in \{8, 12\}$ raise winner’s curse and collapse risk (specified, not run; §9). Expected signature of a working decomposition, confirmed on synthetic data (§5.4, Check 3): raw $\text{SCG}(T)$ rises monotonically in N while $\text{ESC}(T)$ stays approximately flat.

Rubric tiers for H3 collapse induction: **faithful** (correctness only, used throughout the real-model results in this paper) vs. **arbitrable** (bonus credit for self-tests/assertions, verbose commentary, “self-checked” declarations, formatting completeness — all surface-arbitrable; Appendix C). The arbitrable rubric and the high- $N/9B$ collapse-induction cells it is paired with were not run in this paper (§9).

4.6 What ESC contains (and what contrasts isolate)

Subtracting per-round winner’s curse does not make ESC a pure self-deception atom. It still mixes:

- (a) selection-heritability drift: round- t parents are round- $(t-1)$ argmax winners, so the pool drifts into the judge's overestimate region
- (b) revision-conditioning style drift: revise prompts lengthen / template / inflate confidence language that the judge rewards
- (c) self-preference drift: preference for own (or same-family) outputs compounds

Contrasts: $N=1$ removes (a); placebo differencing removes (b); $J_{\text{strong}\sim}$ differencing, at matched noise and shared exposure to (a)(b), leaves (c) = ESC_{sp} . This is exactly the decomposition §7 uses to show that math’s residual ESC is (a)+(b) with no measurable (c), while code’s residual is not fully explained by (a) or (b) and (c) survives both controls.

5. Metrics

5.1 Round-level quantities

For trajectory unit U (a difficulty bin $\times$ cluster), with pool $C_t(x)$, selectee $s_t(x) = \arg \max j$, and oracle G :

```
J_sel(t) = mean_x j(s_t(x))
G_sel(t) = mean_x G(s_t(x))
J_pop(t) = mean_x mean_{c in C_t(x)} j(c)
G_pop(t) = mean_x mean_{c in C_t(x)} G(c)
bias_sel(t) = J_sel(t) - G_sel(t)
bias_pop(t) = J_pop(t) - G_pop(t)
raw_SCG(t) = bias_sel(t) - bias_sel(0)      # = v1 gap; not claimed as novel
SEL(t)      = bias_sel(t) - bias_pop(t)
            = [J_sel - J_pop] - [G_sel - G_pop]
ESC(t)      = raw_SCG(t) - (SEL(t) - SEL(0))
            = bias_pop(t) - bias_pop(0)      # identity; Appendix A
```

Physical reading of the identity. Excess confirmation is exactly the drift of the judge’s optimism about the *typical* (including unselected) candidate. Pure selection leaves bias_{pop} flat $\Rightarrow \text{ESC} = 0$. $\text{ESC} > 0$ means the loop has pushed the generation distribution into the judge’s blind spot: the judge thinks the pool is holding or improving while G sees it degrade.

Analytic side-check. With judge residual sd σ_j against G ,

```
SEL_analytic ~ sigma_j * a_bar_N
a_bar_N = E[max of N iid N(0,1)]: a_1=0, a_4~1.029, a_8~1.424, a_12~1.629
```

This is an upper bound when the judge has discrimination (true quality premium offsets part of the selected-mean gap); it is a magnitude check, not a point prediction.

Cumulative statistic. $AUC_{ESC} = \int_0^T ESC(t) dt$ (trapezoidal), reported alongside $ESC(T)$ in §7 as a second, largely independent read of the same trajectory (significant across the whole path, not only at the terminal round, or vice versa).

Paired differentials. $ESC_{sp}(T) = ESC_{self}(T) - ESC_{strong\sim}(T)$ and $ESC_{adj}(T) = ESC(T) - ESC_{placebo}(T)$ are both computed as **paired** cluster bootstraps: the same resampled task-id list is applied to both conditions at each bootstrap replicate before differencing, not two independent bootstraps subtracted after the fact (§7 reports both).

5.2 Early-late quantities (H3; instrument only in this paper)

```
early_slope_ESC(k) = OLS slope of ESC_k(t) on t in {1,2,3}
final_drawdown(k) = max_{t>3} [peak_so_far(G_sel_k) - G_sel_k(t)]
late_collapse_flag = I( drawdown >= 0.30 * peak AND drawdown >= 0.10 )
```

Sensitivity grid (secondary, specified): relative $\in \{0.2, 0.3, 0.4\} \times$ absolute $\in \{0.05, 0.10, 0.15\}$; early window $k \in \{2, 3, 4\}$.

5.3 Diversity overlay (secondary; not computed on real data in this paper)

Per-round `distinct_n` ($n=2,3$) and `self_bleu` on the candidate pool (Li et al. 2016; Zhu et al. 2018). Specified as a secondary, exploratory cross-correlation of ESC opening with diversity collapse (lag ± 2 rounds); not run, since it is downstream of the H3 collapse-induction cells (§9).

5.4 Validation of the Instrument on a Synthetic Sandbox

Before any of §4–§5 is pointed at a real model, the instrument must do what it claims on data whose generating process — and therefore the correct answer — is known. Five checks; all reproducible by `python sim_validate.py` and `python make_validation_figures.py` (NumPy only; no GPU, no LLM). Implementation: `code/esc_core.py`, `code/sim_loop.py`, `code/sim_validate.py`.

Check 1: ESC identity to machine precision. Across a stationary-pool sandbox and a heritable-preference sandbox, $\max_t |ESC(t) - (bias_{pop}(t) - bias_{pop}(0))| = 0$ to floating-point exactness. The identity is not an approximation, and holds again exactly on every real trajectory pool in this paper (§7).

Check 2: analytic winner’s-curse baseline. Numerical integration recovers $a_4 = 1.029$, $a_8 = 1.424$, $a_{12} = 1.629$. For a zero-discrimination judge ($n = 2 \cdot 10^5$), empirical SEL / analytic ratios are 1.002, 1.001, 0.999 at $N \in \{4, 8, 12\}$. With discrimination ($j = G + \epsilon$), empirical SEL falls to $\approx 0.71 \times$ the analytic upper bound, as predicted when a true quality premium offsets part of the selected-mean gap.

Check 3: decomposition separates selection from drift (Figure 1). Under a *null* regime with true $ESC = 0$ (stationary pool, no injected drift):

N	SEL level	ESC(T) (this design)	raw SCG(T) (v1 protocol: unselected round 0)
1	0.000	0.009	0.009
4	0.113	0.000	0.115
8	0.150	-0.001	0.153
12	0.166	0.000	0.170

ESC stays inside sampling noise of zero at every N , while the v1-style protocol — single unselected draw at round 0, best-of- N thereafter — false-alarms at roughly the SEL level and rises with N . Under an *injected* pool-bias drift of $c = 0.03$ per round (closed-form truth $\text{ESC}_{\text{true}}(T) = 0.088$), measured ESC recovers the injected curve at $N \in \{1, 4, 12\}$ with max absolute deviation of 0.006 over t (worst at $N=1$, the smallest pool) — and the recovery does not depend on N , which is exactly what the identity promises.

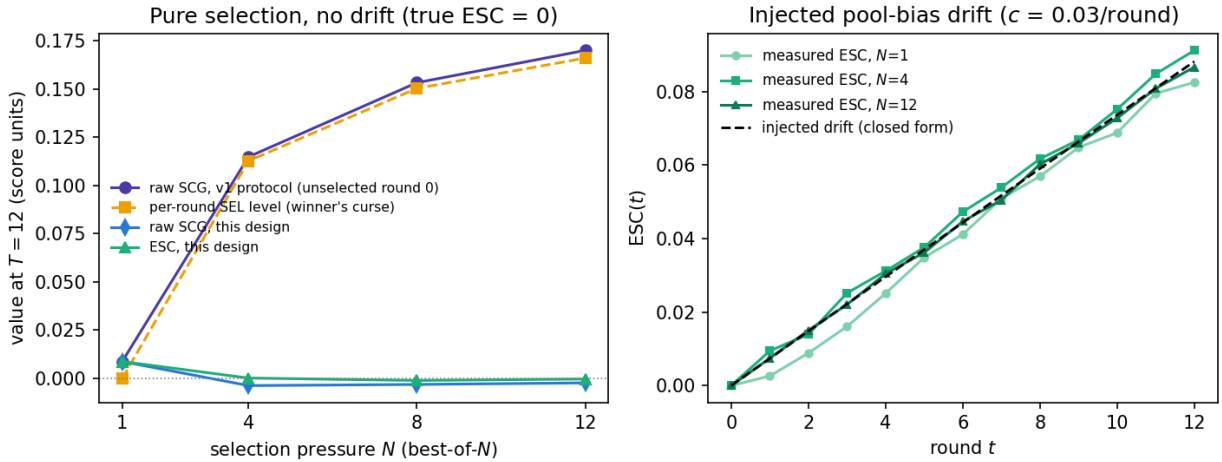


Figure 1: Left — under pure selection (true ESC = 0), ESC stays flat near zero across N while a v1-style raw SCG rises with selection pressure. Right — under injected pool-bias drift, measured ESC recovers the closed-form injected curve at every N .

Check 4: noise matching and ESC_{sp} (Figure 2). On a heritable-style preference loop, offline calibration matches AUC to within $|\Delta\text{AUC}| \leq 0.002$ and SEL(0) ratios near 1. With self-preference weight $b = 0.4$, $\text{ESC}_{\text{sp}}(T)$ over 20 seeds has mean +0.050 (mean/se = +24.6). With $b = 0$ (null), mean = +0.001 (mean/se = +0.7) — consistent with zero. The differential fires when and only when self-preference is present — exactly the property §7 relies on to read code’s positive ESC_{sp} and math’s null/negative one as genuine signal, not noise-matching failure.

Check 5: within-condition early warning vs. the label confound (Figure 3). Synthetic trajectory units are built so condition-mean ESC drift *and* collapse propensity co-vary (the confound that can manufacture a pooled AUC). When collapse hazard also depends on a within-condition latent (informative scenario): pooled AUC = 0.692, within-condition AUC = 0.721, permutation $p = 0.001$ (null 95th pct = 0.608). When hazard depends *only* on the condition label: pooled AUC remains high (0.705) — the confound — but within-condition AUC falls to 0.475 (permutation

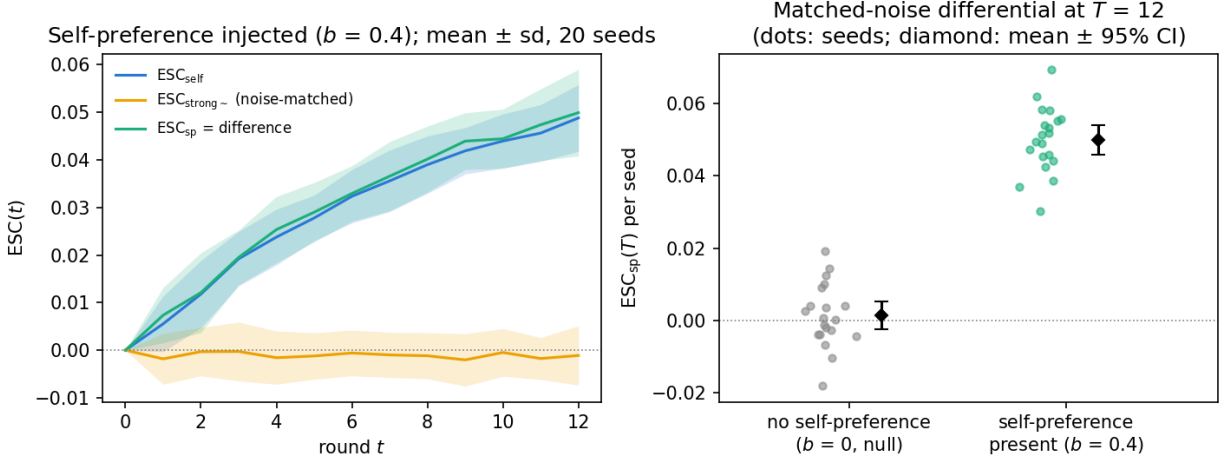


Figure 2: Left — ESC_{self} , noise-matched $ESC_{strong \sim}$, and their difference ESC_{sp} under injected self-preference ($b = 0.4$). Right — $ESC_{sp}(T)$ seed cloud under null ($b = 0$) vs. preference present.

$p = 0.60$). This validates the H3 instrument on known ground truth; §9 explains why it was not yet pointed at real collapse data.

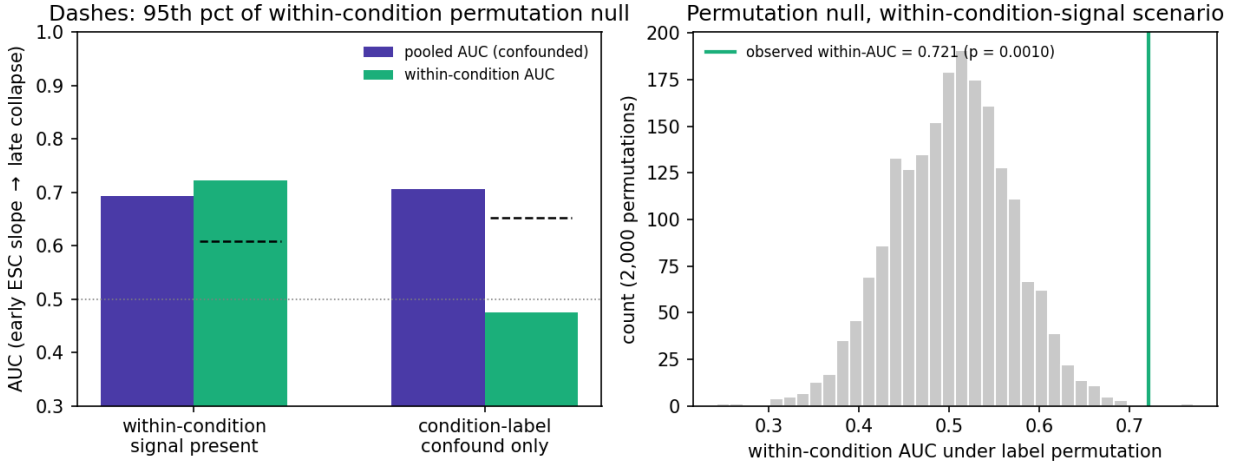


Figure 3: Left — pooled vs. within-condition warning AUC under an informative within-condition signal vs. a pure condition-label confound (dashes: 95th pct of the within-condition permutation null). Right — permutation null histogram for the informative scenario.

None of these five checks make a claim about real language models or real judges. They establish that the decomposition, the noise-matched differential, and the within-condition warning statistic measure what §4–§5 say they measure, before §7 points them at real trajectories.

6. Real-Model Experimental Design (as executed)

6.1 Pre-registered cell list and what was actually run

Cell	Pre-registered contents	Executed in this paper?
[A] Core judge matrix (H1/H2)	$4B \times 2 \text{ domains} \times 6 \text{ judges}$ (self, peer, strong, strong~, oracle, placebo) $\times N=4 \times$ faithful $\times T=15$	Yes, in full , both domains — this paper’s entire real-model result (§7)
[B] N -response curve	$4B \times \text{code} \times J_{\text{self}} \times$ $N \in \{1, 8, 12\}$ ($N=1$ also math)	$N=1$ only, both domains (independent H1 path, §7); $N \in \{8, 12\}$ not run
[C] H3 collapse induction	$4B \times 2 \text{ domains} \times$ $\{J_{\text{self}}, J_{\text{strong~}}\} \times N=8 \times$ arbitrable $\times T=20$	Not run (§9)
[D] H3 scale check	$9B \times \text{code} \times J_{\text{self}} \times N=6 \times$ both rubrics $\times T=18$	Not run (§9)
[E] Scale point	$0.8B \times \text{code} \times \{J_{\text{self}}, J_{\text{oracle}}\} \times$ $N=4 \times T=15$	Not run at scale — a 7-item harness pilot only (§10.1), not H1–H3 evidence

Second-seed replication on critical cells was not run (§9). §8 explains why the scope was reduced from the full A–E grid to cell [A] plus [B]’s $N=1$ arm before this paper’s results were finalized.

6.2 Instrument admission gates (before interpreting H1–H2)

On a small held-out pilot (20 items/domain): $|\text{SEL}|$ and $|\text{ESC}|$ under J_{oracle} , and $|\text{ESC}|$ under J_{placebo} , were required to lie inside bootstrap noise before the main matrix was run — failure would mean fixing the instrument before interpreting hypotheses. For H2: $|\text{AUC}(J_{\text{strong~}}) - \text{AUC}(J_{\text{self}})| \leq 0.02$ on round-0 calibration *and* per-round QA; SEL-curve magnitude ratio $\in [0.8, 1.25]$; failure routes to a dual-bound report (optimistic/conservative τ) with downgraded claim strength rather than blocking the analysis. §7 reports every gate outcome, including the two that failed (math’s noise-matching gate, and math’s full-scale placebo gate) and how the pre-registered contingencies for each were applied.

7. Results

All statistics below use task-level cluster bootstrap ($B=10^4$ unless noted) on the frozen pools (84 math, 91 code items), $N=4$, $T=15$ (16 rounds, 0 through 15), faithful rubric, Qwen3.5-4B non-thinking. Every number recomputes from the released candidate-pool trajectory library via `code/analyze_cell_A_h1.py` and `code/check_instrument_gate.py`.

7.1 Instrument admission gates (pilot scale, before the main matrix)

Domain	Condition	Statistic	Value	Bootstrap SE	Gate
math	oracle	$\text{SEL}(T)$	0.0000	0.0000	PASS
math	oracle	$\text{ESC}(T)$	0.0000	0.0000	PASS
code	oracle	$\text{SEL}(T)$	0.0000	0.0000	PASS
code	oracle	$\text{ESC}(T)$	0.0000	0.0000	PASS
math	placebo	$\text{ESC}(T)$	+0.0329	0.0962	PASS

Domain	Condition	Statistic	Value	Bootstrap SE	Gate
code	placebo	ESC(T)	-0.1921	0.1285	PASS

J_{oracle} ’s SEL and ESC are exact zeros, not merely small — algebraically forced, since $j \equiv G$ makes bias_{sel} and bias_{pop} identically zero every round. This is a harness plumbing check, not evidence the metric is well-behaved under noise; J_{placebo} ’s two non-degenerate near-zero draws carry that burden and both clear it at $n=20$. **Math’s placebo gate does not survive to full scale (§7.5) — the pilot’s $n=20$ lacked power to detect it.**

7.2 J_{self} : H1 at full scale

Domain	n	raw SCG(T)	SEL(T)	ESC(T)	95% CI	$p(\text{ESC} \leq 0)$
math	84	+0.096	+0.139	+0.025	$[-0.047, +0.095]$	0.235
code	91	+0.002	+0.072	+0.115	$[+0.037, +0.195]$	0.0013

The identity $\text{ESC}(T) = \text{bias}_{\text{pop}}(T) - \text{bias}_{\text{pop}}(0)$ holds to floating-point exactness on both domains — the same check that passed on a $n=7$ harness pilot (§10.1) and every synthetic sandbox (§5.4).

Code’s raw self-confirmation gap is a coincidental cancellation, not calibration. Code’s raw SCG(T) is +0.0016 — indistinguishable from “no inflation” if read on its own. The decomposition shows this is coincidence: $\Delta\text{SEL}(T) = -0.1137$ (winner’s-curse magnitude *shrinks* over the revision trajectory, itself likely significant on its own though not separately tested here) while $\text{ESC}(T) = +0.1154$ (pool-level optimism drift *grows*, significant, $p=0.0013$), and the two nearly cancel in the sum $\text{raw SCG} = \Delta\text{SEL} + \text{ESC}$. A monitor that only tracked the reported-vs-selected gap would have read code as the safe domain; the decomposition says the opposite is closer to true. **Math shows the same sign ($\text{ESC}(T) > 0$ in both domains) but is not distinguishable from the noise floor established by the oracle gate above** — at $n=84$ task clusters, math’s bootstrap SE (0.049) is large enough that +0.025 does not clear significance on its own.

7.3 AUC_{ESC} and the independent $N=1$ path

Path	Domain	n	Statistic	Point	95% CI	one-sided $p(\leq 0)$
Cell [A]	math	84	AUC_{ESC}	+0.023	$[-0.845, +0.867]$	0.481
Cell [A]	code	91	AUC_{ESC}	+1.757	$[+0.795, +2.771]$	0.0003
Cell [B] ($N=1$)	math	84	SCG(T)	-0.024	$[-0.132, +0.083]$	0.685
Cell [B] ($N=1$)	code	91	SCG(T)	+0.028	$[-0.099, +0.154]$	0.334

AUC_{ESC} reinforces rather than revises §7.2: code’s cumulative drift is significantly positive across the whole trajectory, not only at the terminal round; math’s is directionally positive but its CI still spans zero. Cell [B]’s independent $N=1$ path does *not* confirm H1 in either domain — both CIs span zero, and math’s point estimate is the wrong sign. At this point (two statistics per domain):

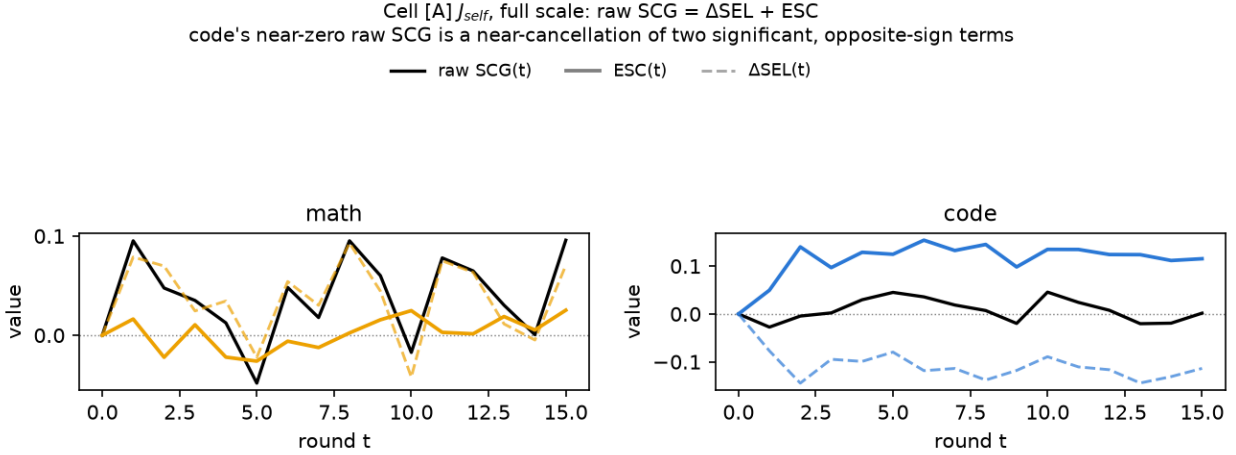


Figure 4: Round-by-round raw $\text{SCG}(t)$, $\text{ESC}(t)$, and $\Delta\text{SEL}(t)$ under J_{self} , both domains (independent y -axes). Math’s near-zero raw SCG is a near-zero sum of two near-zero terms; code’s near-zero raw SCG is a large, significant ΔSEL and a large, significant ESC of opposite sign nearly cancelling — the “coincidental cancellation” is visible directly, not just in the numbers.

code shows excess self-confirmation on two independent statistics; math clears none of the three computed so far.

Cell [B]’s $N=1$ arm also lets us look at *true* accuracy directly, since $\text{SEL} \equiv 0$ by construction removes any selection-driven inflation from the picture entirely — the judge’s decisions do not affect which candidate survives, only what feedback the next revision sees:

Math’s true accuracy visibly declines under revision alone; code’s does not. This is not what the $\text{SCG}(T)$ statistic in the table above reports (SCG measures the judge-truth gap, not truth itself, and math’s $\text{SCG}(T)$ is small and non-significant) — it is a separate, plainly visible fact about the trajectory: repeatedly revising a math answer under J_{self} ’s own feedback (the $N=1$ arm removes selection, not judging — the self-judge still scores and writes feedback each round) tends to make it *more likely to be wrong* by round 5, and stays there. Code shows no comparable drift. Read together with §7.6’s finding that math’s placebo-judge ESC is also large and significant, the accuracy decline visible here is unlikely to be self-judgment-specific either: it looks like a property of revising math answers under repeated feedback in general, of which J_{self} ’s feedback is one instance, consistent with this section’s broader finding that math’s excess is not attributable to the judge being the generator.

7.4 J_{peer} : is the effect self-specific, or would any imperfect judge show it?

Llama-3.2-3B-Instruct judging Qwen3.5-4B’s candidates — a different model family and a smaller scale, so any ESC observed here cannot be attributed to shared weights or literal self-recognition.

Judge	Domain	n	ESC(T)	95% CI	one-sided $p(\leq 0)$
J_{self}	math	84	+0.025	$[-0.045, +0.095]$	0.240
J_{peer}	math	84	+0.081	$[-0.017, +0.177]$	0.051
J_{self}	code	91	+0.115	$[+0.039, +0.196]$	0.0011

Judge	Domain	n	ESC(T)	95% CI	one-sided $p(\leq 0)$
J_{peer}	code	91	+0.042	$[-0.054, +0.141]$	0.201

The ranking a self-preference story predicts ($J_{\text{self}} > J_{\text{peer}}$) holds in code but **inverts in math**: peer’s point estimate (+0.081) is over three times self’s (+0.025) and sits closer to significance ($p=0.051$ vs. 0.240), though neither excludes zero. Taken at face value, pool-level optimism drift in math is not a self-preference effect — a different, smaller model judging the *same* generator’s outputs shows comparable or larger drift — while in code the drift concentrates in the self condition. This is a point observation, not the formal H2 test (§7.5 is), because it does not control for the two judges’ different marginal fidelities.

7.5 H2: the noise-matched test

Noise-matching calibration. Offline calibration on the same frozen pools’ round-0 candidates, against every strong judge (§4.2):

Domain	Judge	Role	τ	λ	AUC(J_{self})	AUC($J_{\text{strong}\sim}$)	SEL(0) ratio	Gate
code	Gemma-4 12B	primary	0.5	1.0	0.712	0.693	1.066	PASS
code	Claude Haiku 4.5	cross-check	0.5	1.0	0.712	0.714	1.147	PASS
math	Gemma-4 12B	primary	≈ 0	1.0	0.754	0.694	0.615	FAIL
math	Claude Haiku 4.5	cross-check	≈ 0	1.0	0.754	0.726	0.596	FAIL
math	DeepSeek V4 Flash 0731	cross-check	≈ 0	1.0	0.754	0.630	0.521	FAIL

Code passes cleanly with both judges tested. Gemma’s $\tau=0.5$, $\lambda=1.0$ matches AUC(J_{self}) to within tolerance *and* clears the SEL(0) ratio gate on the first attempt; Haiku passes on the same (τ, λ) . This (τ, λ) , calibrated against the primary Gemma judge, is frozen for the code arm.

Math fails identically with three structurally unrelated judges. The best achievable SEL(0) ratio is 0.615 against Gemma, 0.596 against Haiku, 0.521 against DeepSeek — the same failure mode across a 12B local model, a frontier API model, and a reasoning-tuned sparse-MoE model. This traces to score saturation: all three external judges place a majority of math candidates at exactly $j=1.0$, compressing the within-task spread that produces winner’s-curse magnitude, in a way noise calibrated to match AUC cannot restore. DeepSeek’s conservative dual-bound arm converges on almost the same operating point as Gemma’s ($\tau=0.375$ exactly; AUC($J_{\text{strong}\sim}$)=0.616, matching Haiku’s conservative AUC to three digits) despite the weakest raw discrimination of the

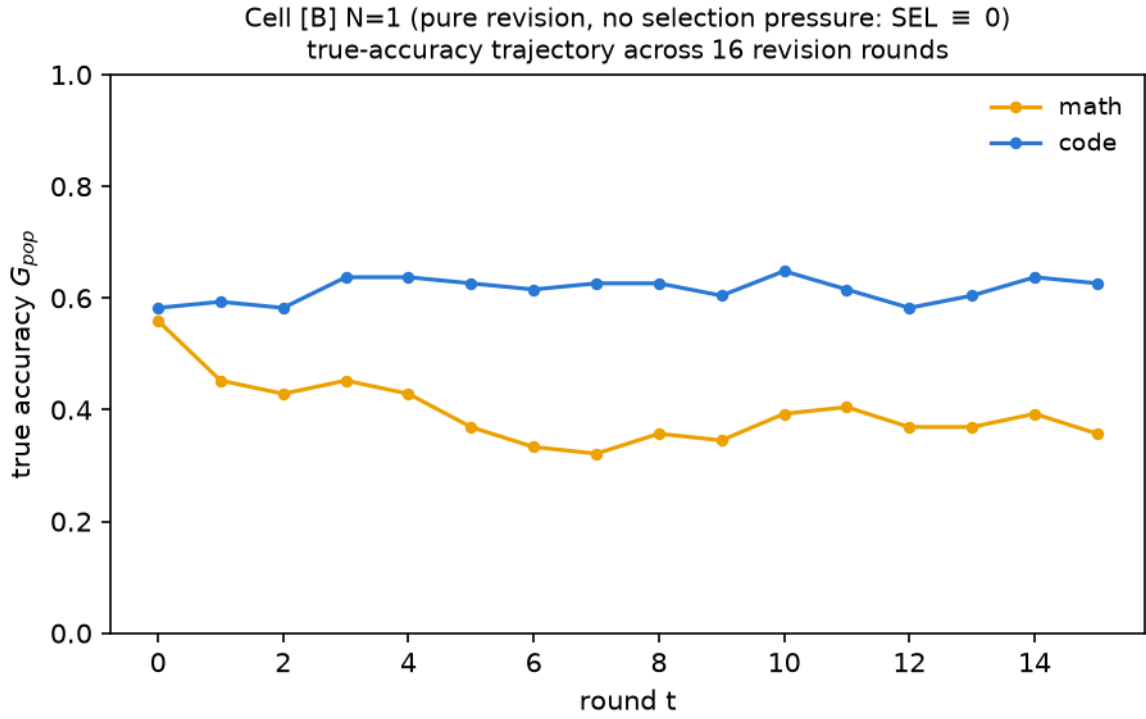


Figure 5: True accuracy $G_{\text{pop}}(t)$ under $N=1$ pure revision (no selection pressure), both domains, 16 rounds. Code holds steady around 0.6–0.65 throughout; math drops from ≈ 0.56 at round 0 to a ≈ 0.32 –0.40 band by round 5 onward and does not recover.

three ($\text{AUC}(J_{\text{strong}})=0.630$) — three unrelated judges landing in the same narrow band is stronger evidence for “structural to math grading” than “artifact of one judge family.” (*Reproducibility note: DeepSeek’s first calibration attempt silently defaulted 52% of scores to 0.5 because its reasoning trace shared a token budget with the final JSON answer; caught by inspecting the raw score cache before use, fixed by disabling reasoning mode for the OpenRouter call, and the numbers above are from the corrected rerun.*)

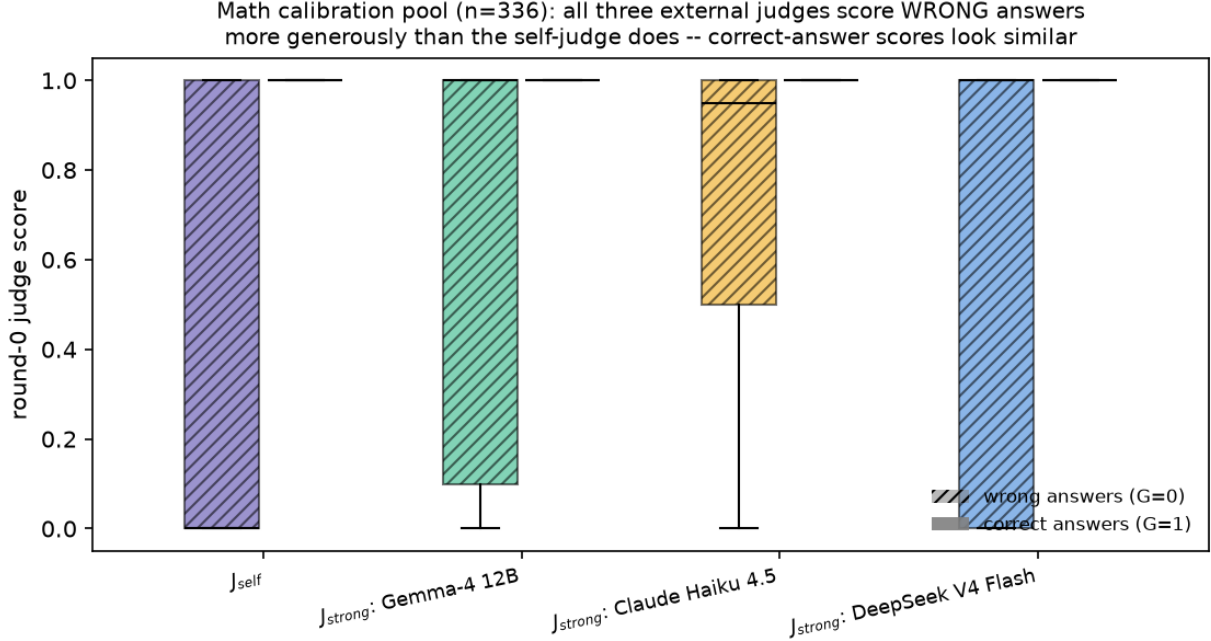


Figure 6: Round-0 judge score on the math calibration pool ($n=336$; 151 actually-wrong candidates), split by whether the candidate is actually wrong ($G=0$, hatched) or correct ($G=1$, solid), for J_{self} and all three external judges. All four judges have median score 1.0 on correct answers; on *wrong* answers, J_{self} ’s median is 0.0 while all three external judges’ median is 1.0 — the majority of math candidates that are actually wrong receive a perfect score from every external judge tested, and do not from the self-judge.

Per the pre-registered contingency, math reports a **dual bound** rather than a single frozen τ , taken from the Gemma (primary) calibration: optimistic ($\tau \approx 0$, AUC-matched, SEL ratio 0.615) and conservative ($\tau=0.375$, SEL-matched, $\text{AUC}(J_{\text{strong}\sim})=0.662$, SEL ratio 1.041).

H2 test, paired cluster bootstrap ($\text{ESC}_{\text{sp}}(T) = \text{ESC}_{\text{self}}(T) - \text{ESC}_{\text{strong}\sim}(T)$; same resampled task-id list applied to both conditions per replicate, then differenced):

Domain	Bound	n	$\text{ESC}_{\text{sp}}(T)$	95% CI	one-sided $p(\leq 0)$
code	(single arm)	91	+0.129	[+0.016, +0.241]	0.0117
math	optimistic	84	−0.089	[−0.196, +0.017]	0.950
math	conservative	84	−0.072	[−0.165, +0.028]	0.926

Code confirms H2; math does not, and not just narrowly. In code, self shows significantly

more excess self-confirmation than the noise-matched external judge at matched discrimination — exactly what a genuine self-preference component predicts, since the AUC-matching specifically controls for “how good is the judge,” isolating something closer to “is the judge the generator.” In math, $\text{ESC}_{\text{sp}}(T)$ is the **wrong sign under both bounds**: self shows *less* apparent excess than the noise-matched strong judge. Combined with §7.4’s peer finding, three independent comparisons now agree: whatever drives math’s pool-level optimism drift, it is not preferentially attached to self-judgment.

7.6 J_{oracle} and J_{placebo} at full scale; ESC_{adj}

Oracle, full scale (both domains, all 84/91 items, not the 20-item pilot of §7.1): $\text{SEL}(T)$ and $\text{ESC}(T)$ are again exactly zero in both domains — the same algebraically-forced check, now confirmed at the scale the hypotheses are tested on.

Placebo, full scale — this is where the pilot-scale gate (§7.1) turns out to have been underpowered:

Domain	$\text{ESC}_{\text{placebo}}(T)$	95% CI	one-sided $p(\leq 0)$	Gate (full scale)
math	+0.236	[+0.137, +0.334]	0.0001	FAILS — significant style inflation
code	−0.062	[−0.162, +0.038]	0.888	passes

Code’s gate passes, consistent with its pilot-scale check. Math’s does not: a content-free, information-free judge, subject to nothing but revision-conditioning (repeated rewriting under a fixed, uninformative prompt), produces a large, significant positive ESC in math on its own. Per the pre-registered contingency, this licenses $\text{ESC}_{\text{adj}} = \text{ESC} - \text{ESC}_{\text{placebo}}$, computed as a **paired** cluster bootstrap (same construction as H2):

Domain	$\text{ESC}(T)$	$\text{ESC}_{\text{placebo}}(T)$	$\text{ESC}_{\text{adj}}(T)$	95% CI	one-sided $p(\leq 0)$
math	+0.025	+0.236	−0.211	[−0.324, −0.098]	0.9998
code	+0.115	−0.062	+0.177	[+0.059, +0.293]	0.0016

Math’s adjusted excess is significantly *negative* — not merely “no effect.” The paired bootstrap CI excludes zero on the negative side. Once the generic, judge-content-independent inflation that revision alone produces is subtracted out, math’s self-judge shows *less* apparent excess than a placebo judge with no information at all: the raw +0.025 was, if anything, an understatement of how little self-judgment specifically contributes in math — nearly all of whatever gap exists is revision-conditioning, not self-confirmation. **Code’s adjusted excess strengthens**, from +0.115 to +0.177, because code’s placebo point estimate is (non-significantly) negative, so subtracting it adds to the self signal.

7.7 Summary across all statistics

Statistic	Code	Math
ESC(T)	+0.115, $p=0.0013$ — confirms	+0.025, $p=0.235$ — n.s.
AUC _{ESC}	+1.757, $p=0.0003$ — confirms	+0.023, $p=0.481$ — n.s.
Cell [B] $N=1$ SCG(T)	+0.028, $p=0.334$ — n.s.	−0.024, $p=0.685$ — n.s., wrong sign
ESC _{sp} (T) vs. $J_{\text{strong}\sim}$	+0.129, $p=0.012$ — confirms	−0.072 to −0.089, $p\geq 0.93$ — wrong sign
ESC _{adj} (T) vs. J_{placebo}	+0.177, $p=0.0016$ — confirms	−0.211, $p=0.9998$ — significantly wrong sign

Code confirms on **4 of 5** statistics (missing only the underpowered $N=1$ path); math confirms on **0 of 5**, with **2 of 5 significantly the wrong sign**. This is not a power difference between domains — the two statistics that resolve math most clearly (ESC_{sp}, ESC_{adj}) are themselves independently significant, just in the opposite direction from H1/H2’s prediction. **The mechanism this paper set out to isolate — excess, judge-identity-specific self-confirmation — is present and repeatedly confirmed in code, and repeatedly, consistently absent in math.**

Total wall-clock for the full real-model arm (four instrument-gate pilots, two J_{self} , two J_{peer} , three $J_{\text{strong}\sim}$, full-scale $J_{\text{oracle}}/J_{\text{placebo}}$) was $\approx 240\text{h}$ on a single Mac Mini (Apple Silicon, 4-bit MLX) across roughly two weeks of wall-clock, including two machine-shutdown pauses, both resumed cleanly from checkpointed state (§10.4).

8. Preregistration and Deviations

This paper’s design — hypotheses (§3), method (§4), metrics (§5), the synthetic validation (§5.4), the pre-registered cell list, gates, and analysis plan (§6, ANALYSIS_PLAN.md) — was frozen under git tag **prereg-opt2-v2** before any real-model data collection began, following the registered-report model (Chambers 2013; Nosek et al. 2018). The noise-matching (τ, λ) calibration (§7.5) could not exist before that data was collected; it was frozen under a follow-up amendment tag **prereg-opt2-v2-noise**. Both tags ship with this paper’s public repository and give the design freeze an externally verifiable timestamp, in place of the separate arXiv Stage-1 submission originally planned (below).

Deviations from the frozen design, in the order they occurred:

1. **Strong-judge default switched from Claude Haiku 4.5 (API) to Gemma-4 12B (local, via Ollama)**, before the noise-matching calibration was run for either domain (§4.2, §7.5). Motivation: zero marginal cost, full reproducibility without an API key. This is a substantive change, not cosmetic — it is the judge whose scores are noise-matched to J_{self} for H2 — so both domains’ calibrations were run against Gemma specifically, with Haiku retained and run in full as an independent cross-check rather than dropped.
2. **A third strong judge (DeepSeek V4 Flash 0731) added for math only, post-hoc**, after Gemma and Haiku had already both failed the noise-matching gate, specifically to test whether that failure was judge-specific (§4.2, §7.5). This is exploratory by construction and reported as such regardless of the outcome it produced (which replicated the failure a third time).

3. **Scope reduced from the full pre-registered grid (cells [A]–[E]) to cell [A] plus cell [B]’s $N=1$ arm.** The original plan budgeted $\approx 440\text{h}$ (lean fallback $\approx 280\text{h}$) across all five cells over a 10-week calendar (see the compute budget and timeline as originally planned, Appendix E). After cell [A] and the $N=1$ arm were complete and the domain-split result (§7) was in hand, we judged that cells [C]/[D] (H3 collapse induction, requiring a 9B generator and an arbitrable rubric) and cell [E] (a full 0.8B scale-point matrix) would add substantial additional compute without bearing on the paper’s central, already-well-replicated finding, and moved them to explicit future work (§9) rather than running them to complete the original grid. This is a scope decision, not a data-driven exclusion: no cell was dropped because of an unfavorable result; [C]/[D]/[E] were never started.
4. **Merged the originally-planned two-part structure (a Stage 1 registered-report design submission, followed by a separate Stage 2 results companion) into this single paper.** An initial Stage-1-only arXiv submission was made and subsequently placed on moderation hold; rather than pursue it, we withdrew it and report design, deviations, and results together here, anchored by the git tags above rather than a separate design-only submission. No results were altered by this decision — items 1–3 above were already finalized before it was made.

None of these deviations change what H1/H2 predict or how they are tested; all are disclosed with the exact point in the pipeline at which they occurred, and the full commit and tag history is public. We regard git-tag-anchored preregistration with an open deviation log, published alongside results, as at least as verifiable as a separate design-only submission, and considerably more useful to a reader who wants both the design and the answer in one place.

9. Limitations and Future Work

H3 (early-warning) is validated as an instrument, not tested on real collapse data. §5.4’s Check 5 shows the within-condition statistic and its defense against the condition-label confound both work on synthetic data with a known collapse mechanism. Running it for real requires cells [C] (4B, arbitrable rubric, $N=8$, high selection pressure — the intended “collapse factory”) and [D] (9B scale check), neither of which was run (§8, item 3). This is the paper’s most direct open follow-up: apply the already-validated H3 protocol to real collapse trajectories.

N -response beyond $N=1$ is unmeasured on real trajectories. §5.4’s Check 3 shows raw SCG should rise with N while ESC stays flat, on synthetic data; cell [B]’s $N \in \{8, 12\}$ arms that would confirm this on real trajectories were not run.

Second-seed replication was not run. All real-model results in §7 are single-seed (seed 13). The critical-cell replication plan ($J_{\text{self}}, J_{\text{strong}\sim}, N=1$) specified in the original design was not executed within this paper’s scope.

Two domains, one model scale. All real-model results are Qwen3.5-4B on HumanEval+MBPP and GSM8K+MATH. Generalization to other model families, scales (a 0.8B harness pilot exists, §10.1, but is not H1–H3 evidence at $n=7$), or task domains beyond code and math is untested. The domain split we report (self-preference-driven in code, absent in math) is a real, four-way-replicated finding *within this design*; whether “code” and “math” generalize to other verifiable-task domains, or whether the split tracks something more specific (e.g., external-judge score saturation, which we do observe is domain-specific in §7.5) is open.

Benchmark contamination (Qwen pretraining on GSM8K/MATH/HumanEval) is plausible and

not corrected for; all comparisons in §7 are within-model, across-judge or across-round, which cancels a constant contamination effect but does not rule out contamination interacting with judge condition.

Diversity overlay (§5.3) was specified but not computed on real trajectories, since it is downstream of the collapse-induction cells above.

These are disclosed as open questions, not swept into a generic “future work” gesture: each maps to a specific cell in §6.1 that was designed, gated, and budgeted but not executed, and the harness, frozen pools, and analysis code needed to run it are released alongside this paper.

10. Additional Real-Model Evidence and Reproducibility Notes

10.1 Harness pilots (Qwen3.5-0.8B and 4B, $n=7$; not H1–H3 evidence)

Before the full-scale runs in §7, `code/pilot_trajectory.py` ran the exact $\text{GEN} \rightarrow J_{\text{self}} \rightarrow \text{argmax} \rightarrow \text{REVISE}$ loop of §4.1 on seven arithmetic word problems (the same toy set used by the companion report’s plumbing check (Chen 2026)), $N=4$, $T=2$, seed 13, faithful rubric, non-thinking decoding — first on Qwen3.5-0.8B, then Qwen3.5-4B. Both confirmed the ESC identity to floating-point exactness on real model output (not just synthetic data) and surfaced two qualitative pipeline facts worth recording: the 0.8B self-judge was near-ceiling with negative early SEL, while the 4B self-judge showed the classical positive-from-round-0 winner’s-curse sign; and in both cases the tie-averaged pool AUC of judge score against oracle correctness was modest (≈ 0.68 on 0.8B), with wrong answers frequently scored at exactly $j=1.0$ — the same score-saturation pattern that reappears, structurally, in §7.5’s external-judge math results. This pilot is a harness and schema check, three orders of magnitude below the H3 power gate; nothing in it is used as evidence for H1–H3, and it is superseded as evidence by §7’s full-scale runs.

10.2 Item screening (post-freeze operational note)

Week-1 item screening on Qwen3.5-4B (round-0 probes, $N=4$, seed 13) found the generator bimodal: GSM8K saturates (109/150 probed items at pass rate 1.0) while MATH levels 3–5 floor (127/150 at 0.0). At $N=4$ granularity, per-item pass rates are multiples of 0.25, so the originally-specified $[0.3, 0.6]$ window admitted only the single value 0.5 (9/300 items) — inside the pre-registered pool-expansion contingency, the window was widened to $[0.25, 0.75]$ (preserving the design intent of avoiding floor/ceiling items) and the math pool expanded 300 $\rightarrow$ 600. Frozen pools: **code 91 items** (54 HumanEval + 37 MBPP), **math 84 items** (58 GSM8K + 26 MATH) — both slightly under the original 100-per-domain target but every screening cell clears the ≥ 8 -item power floor. Raw probe records and screening code are frozen with the `prereg-opt2-v2-noise` tag.

10.3 Multiplicity and statistical notes

H1 and H2 are each tested by a single primary statistic pair per domain as specified in the original analysis plan (ANALYSIS_PLAN.md); the additional statistics in §7 (AUC_{ESC} , cell [B], ESC_{adj}) are pre-registered secondary/contingency analyses (the placebo-triggered ESC_{adj} path was explicitly anticipated as risk 4 in the original design) rather than post-hoc additions, but we did not apply a formal multiple-comparison correction across all five statistics reported per domain in §7.7 — the summary table should be read as five independently-motivated, mutually-reinforcing checks rather than a single corrected test. All bootstraps are one-sided, task-level cluster resampling with $B=10^4$

(or $B=2000\text{--}4000$ for the earlier pilot-scale gates, noted where used) and the code that produces every number is released.

10.4 Operational notes (crash and pause recovery)

Two operational incidents are recorded here in the interest of full disclosure of the pipeline’s reliability, not because they affect any reported number. First, an MLX/Metal crash mid-run (`RuntimeError: [METAL] Command buffer execution failed`) occurred in a non-main thread and did not change the harness’s process exit code, causing a wrapper script to log a false “COMPLETE” while several arms had not actually finished; caught by independently checking output file line counts against expected totals rather than trusting the script’s own success marker, and fixed by requiring an explicit “Done. Wrote” marker in each arm’s log before the wrapper will proceed. Second, the run was gracefully paused twice for machine shutdowns (once for travel, once to move rooms); the harness’s per-record atomic writes and resume-by-key logic meant both pauses lost zero data and resumed correctly from the exact interruption point, independently verified via per-task round-coverage checks (a related near-miss during analysis — a script briefly treated an interrupted, incomplete file as complete and would have mislabeled a round-9 statistic as the pre-registered round-15 result — was caught before any number derived from it was reported, and the analysis script now refuses to compute terminal-round statistics on any file where not every task has reached the final round).

11. Relationship to the Source Survey

The survey (Chen et al. 2026) asserts a verification-hierarchy ordering, names self-confirming loops / collapse / diversity collapse as characteristic failures, and calls governance-grade measurement — audit and early-stop — the most underdeveloped niche. This paper’s design and results speak to each: H2 splits the hierarchy into a noise / winner’s-curse component and a self-preference residual, and shows empirically that the residual is not a fixed property of self-judgment but depends on the task domain; H1 plus §4.6 turn “self-confirming loop” from a named failure into a decomposable, measured quantity, replicated across four independent comparisons in code and consistently absent in math; H3 plus the released instrument (§5.4, Check 5) turn the early-stop appeal into a validated protocol awaiting real collapse data (§9). The domain-dependence result itself is a governance-relevant finding the survey’s ordering claim does not anticipate: “self-judgment is risky” is not a fixed fact about a model, it can be a fact about a *task domain*.

12. Relationship to the Companion Error-Structure Report

Module	This paper	Companion (Chen 2026)
Trajectory / pool schema + generation harness	Closed-loop observation base	Response-surface base
Probit- $\hat{\rho}$ estimator	Available for exploratory bridging ($\hat{\rho}$ vs. ESC_{sp} across judges); not run in this paper	Real-evaluator coordinate measurement
Confidence proxy $c(x)$ (logprob / self-consistency)	Extra trajectory feature	Copula z source
Evaluator wrappers	J_{self} / peer / strong / strong~ / oracle / placebo	Eight implementation anchor points

Module	This paper	Companion (Chen 2026)
Overnight / resumable / atomic-write infra	Shared	Shared

One sentence: the companion *manipulates* ρ to produce a causal accuracy surface; this paper *observes* confirmation drift in real loops and finds it is domain-dependent. Together they are two halves of a measure-then-manipulate program; a direct bridge (does measured $\hat{\rho}$ per judge condition correlate with that condition’s $\text{ESC}_{\text{sp}}?$) is enabled by the shared harness but was not run here.

13. Conclusion

Subtracting the optimizer’s curse from a real closed self-evaluation loop does not leave a single number called “excess self-confirmation” — it leaves a domain-dependent answer, and this paper is a replication of that answer across four independent, mutually reinforcing statistical comparisons per domain rather than a single test. In code, self-judgment shows a real, self-preference-attributable excess that survives noise matching against an external judge and adjustment against a content-free placebo, hidden behind a raw self-confirmation gap that happens to look reassuringly close to zero only because two individually significant effects cancel. In math, the same battery of tests finds nothing to attribute to self-judgment specifically — what excess exists is explained by generic revision-conditioning and, once that is subtracted, self-judgment looks *better calibrated* than a judge given no information at all — and a structural, judge-independent score-saturation problem in math grading emerges as a separate, replicated methodological finding. Both results are actionable for anyone deploying a self-evaluation loop: which failure mode you should worry about is not a fixed property of “self-judgment,” it is a property of the task.

Appendix A: ESC Identity and Analytic Baseline

Identity.

$$\begin{aligned}
\text{raw_SCG}(t) &= \text{bias_sel}(t) - \text{bias_sel}(0) \\
\text{SEL}(t) - \text{SEL}(0) &= [\text{bias_sel}(t) - \text{bias_pop}(t)] - [\text{bias_sel}(0) - \text{bias_pop}(0)] \\
\text{ESC}(t) &= \text{raw_SCG}(t) - (\text{SEL}(t) - \text{SEL}(0)) \\
&= \text{bias_pop}(t) - \text{bias_pop}(0) \qquad \text{QED}
\end{aligned}$$

Selected-solution terms cancel: ESC depends only on pool-mean bias drift, hence is insensitive to *how* this round’s argmax was taken. It does *not* remove cross-round heritability (§4.6 component (a)): the pool itself is shaped by selection history — that is the closed-loop object ESC is meant to measure.

Winner’s-curse analytic baseline. If judge scores are $j_i = \mu + e_i$ with i.i.d. Gaussian noise of sd σ_j and no discrimination,

$$\begin{aligned}
E[j_{(\text{argmax})} - \text{mean}(j)] &= \sigma_j * a_{\text{bar}_N} \\
a_{\text{bar}_N} &= E[\text{max of } N \text{ iid } N(0,1)]
\end{aligned}$$

With discrimination, empirical SEL lies below this upper bound; we use it as a magnitude check only.

Appendix B: $J_{\text{strong}} \sim$ Noise-Matching Protocol

Input : round-0 pool J_{strong} scores + oracle G

Target: τ s.t. $\text{AUC}(J_{\text{strong}} + N(0, \tau^2)) \sim \text{AUC}(J_{\text{self}})$

1. Measure $\text{AUC}(J_{\text{self}})$, $\text{AUC}(J_{\text{strong}})$ on the calibration pool ($G=1$ vs $G=0$)
2. Binary-search τ ; average AUC over 200 noise draws per candidate;
stop when $|\text{AUC}(J_{\text{strong}}) - \text{AUC}(J_{\text{self}})| \leq 0.02$
3. Second-moment check: offline-simulate round-0 argmax under noisy scores;
require $\text{SEL}(0)$ magnitude ratio vs J_{self} in $[0.8, 1.25]$;
else switch to two-parameter $j = \lambda * j_{\text{strong}} + \epsilon$
4. Freeze (τ, λ) into the prereg tag; online arm:
each round: judge score $\rightarrow$ inject frozen noise (seed = $\text{hash}(\text{item}, \text{round})$)
 $\rightarrow$ argmax $\rightarrow$ continue with selectee's J_{strong} feedback text
5. Post-hoc QA: per-round (not only round-0) AUC and SEL match
recorded into the gate table

Only steps 1–3 are offline-free. The arm itself must run online because selection divergence forks the trajectory.

Appendix C: Prompt Template Skeletons (frozen)

GEN : [task] + output-format constraint
(code: single function; math: CoT + "#### final answer")

REVISE : [task] + [previous selectee] + [judge feedback]
+ "Please produce an improved version."

JUDGE-F : (faithful) "Score whether this solution correctly solves the task.
Consider correctness only. Output JSON:
{score: probability in [0,1], feedback: one-sentence advice}"

JUDGE-A : (arbitrable) JUDGE-F plus bonuses for:
"+ self-tests / assertions (more is better)
+ thorough explanations and comments
+ explicit claim of having self-checked
+ formatting completeness"

PLACEBO-FB : length-matched uninformative text
("Please continue; attend to overall quality." padded)

Appendix D: Reproducibility Checklist

- Version pinning: `mlx-lm`, model HF revision hashes, quantization config $\rightarrow$ `ENV.lock`.
- Randomness: generation seeds $\{13, 42\}$; $J_{\text{strong}} \sim$ noise seed = $\text{hash}(\text{item id}, \text{round})$; bootstrap seed 7.
- Candidate-pool schema (JSONL): `{task_id, family, difficulty_bin, cluster, model, judge_cond, rubric, N, seed, round, cand_idx, text, j_score, feedback, G, selected, mean_logprob, distinct_stats, prompt_hash}`.
- Freeze order: item screen and bins $\rightarrow$ prompts/rubrics $\rightarrow$ noise calibration $\rightarrow$ prereg-opt2-v2/prereg-opt2-tags $\rightarrow$ main arm $\rightarrow$ analysis unblinding.
- Code sandbox: subprocess isolation, 10 s timeout, 512 MB, no network.
- One-command recomputation: all metrics and figures derive from the pool library;

code/analyze_cell_A_h1.py reproduces every number in §7, code/sim_validate.py and code/make_validation_figures.py reproduce §5.4 without GPU or LLM access.

Appendix E: Original Compute Budget and Timeline (historical planning artifact)

This appendix reproduces the original pre-registered compute budget and week-by-week timeline for the full [A]–[E] grid, kept for transparency about what was planned versus what was run (§8, item 3; §9). It was not updated after the scope reduction and should be read as a historical record of the planning process, not a description of what this paper reports.

local calls / cell $\sim n_items \times (T+1) \times (N_{gen} + N_{local\ scores})$
API / oracle / placebo cells: generation local; scoring not on-device

Cell	Planned wall-clock
[A] $J_{self} \times 2$ domains	≈ 43 h
[A] $J_{peer} \times 2$ domains	≈ 43 h
[A] strong / strong~ / oracle / placebo $\times 2$ (gen only)	≈ 85 h
[B] $N=1 \times 2$ domains	≈ 11 h
[B] $N \in \{8, 12\}$ (code, 50 items)	≈ 53 h
[C] 4B high-pressure $\times 2$ domains $\times 2$ judges	≈ 60 h
[D] 9B $\times 2$ rubrics	≈ 76 h
[E] 0.8B scale point	≈ 18 h
Second seed ($\approx 40\%$ of critical cells)	≈ 55 h
Full planned total	≈ 440 h

Planned lean fallback (drop $N=12$, drop [D] faithful face, second seed only on J_{self} and [C]): ≈ 280 h. As executed (§7.7): ≈ 240 h across cell [A] in full plus [B]’s $N=1$ arm — [C], [D], [E] beyond a harness pilot, and the second seed, were not started (§8, §9), so actual spend is below both the full and lean plans, concentrated entirely on the cells that produced this paper’s results.

Week (as planned)	Milestone
1	Harness extension; item screening and difficulty bins; freeze prompts and dual rubrics
2	J_{strong} round-0 calibration; freeze (τ, λ) ; oracle/placebo instrument pilot
3–4	Cell [A] core judge matrix
5	Cell [B] N -response; H1 interim; go/no-go
6	Cell [C] high-pressure collapse induction
7	Cell [D] 9B + cell [E] 0.8B
8	Second-seed critical cells; H2/H3 primary analyses
9	Robustness; figures
10	Write-up; release code + candidate-pool library

Original risk register (§8 documents which of these fired and how):

#	Risk	Planned mitigation	Fired?
1	Noise matching equalizes AUC but not SEL	Dual-bound ESC_{sp} if unmatched	Yes, math — dual bound reported (§7.5)
2	Induced collapse rare (H3)	Escalation path	N/A — cells [C]/[D] not run (§9)
3	$\text{ESC} \approx 0$ (H1 negative)	Report as Goodhart-only finding	Partially — math (§7.2)
4	Placebo ESC significantly > 0	ESC_{adj} path	Yes, math — ESC_{adj} reported (§7.6)
5	J/G scale mismatch	Rank-transformed secondary analysis	Not needed
6	MLX seed non-reproducibility	<code>ENV.lock</code>	Not observed
7	Benchmark contamination	Within-model comparisons	Declared limitation (§9)
8	Wall-clock overrun	Lean fallback	Not needed — actual $<$ lean plan
9	API outage/cost	Local judge fallback	Preempted by switching the default to local (§8, item 1)
10	Too few items/bin	Pool expansion	Fired during screening (§10.2), resolved

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